

Notes: Allen and Gale (JPE, 2000)

Financial Contagion

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1 One-sentence summary

Allen and Gale build a microeconomic theory of financial contagion in which banks use interregional claims to share liquidity risk, but the same claims can transmit a small liquidity shock from one region to the whole financial system when the interbank network is incomplete and sufficiently connected.

2 Big picture

- **Question.** How can a small liquidity shock in one part of the banking system spread into an economywide financial crisis?
- **Core mechanism.** Banks hold claims on banks in other regions to insure against regional liquidity-preference shocks. These cross-claims are useful in normal states, but they create balance-sheet spillovers in crisis states.
- **Key tension.** Interbank linkages improve risk sharing ex ante but can propagate losses ex post.
- **Main result.** A complete interbank claims structure is more robust than an incomplete connected structure. In a complete structure, losses from one region are spread thinly. In an incomplete connected structure, losses hit neighboring regions hard enough to start a chain reaction.
- **Network insight.** It is not simply that more links are always more dangerous. The danger comes from the combination of incomplete insurance, connected chains, and the private incentive to hoard liquidity.
- **Policy insight.** A central bank can be interpreted as an institution that helps complete markets and prevents inefficient liquidation chains.

3 Incremental contribution of the paper

3.1 A microfoundation for financial contagion

The paper provides a formal equilibrium model in which contagion arises from real balance-sheet linkages between banks, not from vague panic or arbitrary sunspots. The mechanism is based on overlapping claims among banks that were originally created for efficient liquidity insurance.

3.2 Contagion as an essential equilibrium phenomenon

A central contribution is that under certain conditions contagion is not merely one possible equilibrium. In the key incomplete-market case, every continuation equilibrium after the perturbation involves bankruptcy in all regions. This makes the contagion result stronger than a self-fulfilling-bank-run story.

3.3 Network structure matters for robustness

The paper shows that the same first-best risk-sharing allocation can be decentralized by different interbank networks, but those networks have very different crisis properties. Complete networks, connected incomplete networks, and disconnected incomplete networks can all work in normal times, but they differ sharply in crisis states.

3.4 The free-rider logic of liquidity provision

The paper clarifies why incomplete networks can be fragile. When there is an aggregate shortage of liquidity, the socially efficient response is for many regions to liquidate a small amount of long assets. But each region prefers to avoid costly liquidation and instead withdraw claims on others. This pecking-order behavior can push the crisis onto neighboring banks.

3.5 A benchmark for later network-contagion research

The model predates and motivates much later work on financial networks. Its main lesson is that the architecture of claims determines whether interbank linkages provide insurance or transmit distress.

4 Model setup

4.1 Timing and agents

There are three dates, $t = 0, 1, 2$. Consumers are ex ante identical and endowed with one unit of the consumption good at date 0. At date 1 they learn whether they are early or late consumers.

Consumer preferences follow the Diamond-Dybvig structure:

$$U(c_1, c_2) = \begin{cases} u(c_1), & \text{with probability } \omega, \\ u(c_2), & \text{with probability } 1 - \omega. \end{cases} \quad (1)$$

Early consumers value only date-1 consumption. Late consumers value only date-2 consumption. A consumer's type is private information, so late consumers can imitate early consumers.

4.2 Assets

There are two assets.

- **Short asset.** One unit invested at date t produces one unit at date $t + 1$.
- **Long asset.** One unit invested at date 0 produces $R > 1$ at date 2. If liquidated early at date 1, it produces only $0 < r < 1$.

The long asset is productive if held to maturity but costly to liquidate early. This liquidation cost is central to the crisis mechanism.

4.3 Regions and liquidity shocks

The economy has four ex ante identical regions, A, B, C, D . The probability that a consumer is early differs across regions and states. Let ω_H and ω_L be the high and low liquidity-demand shocks, with

$$0 < \omega_L < \omega_H < 1. \quad (2)$$

Define the average fraction of early consumers as

$$\gamma = \frac{\omega_H + \omega_L}{2}. \quad (3)$$

In normal states, exactly two regions have high liquidity demand and two have low liquidity demand, so aggregate liquidity demand is constant.

4.4 Banks

Banks collect deposits, invest in the short and long assets, and offer deposit contracts. In a decentralized equilibrium, a representative bank in each region offers a standard deposit contract (c_1, c_2) : depositors can withdraw c_1 at date 1 or c_2 at date 2.

Banks also hold interbank claims. These claims allow banks in high-liquidity-demand regions to obtain resources from banks in low-liquidity-demand regions in normal states.

5 First-best risk sharing

5.1 Planner's problem

A planner chooses investment in the long and short assets, denoted (x, y) , and consumption (c_1, c_2) . The feasibility constraints are

$$x + y \leq 1, \tag{4}$$

$$\gamma c_1 \leq y, \tag{5}$$

$$(1 - \gamma)c_2 \leq Rx. \tag{6}$$

The objective is expected utility:

$$\gamma u(c_1) + (1 - \gamma)u(c_2). \tag{7}$$

The first-best allocation is the unique solution to this problem.

5.2 Incentive compatibility

Because late consumers can mimic early consumers, the allocation must satisfy

$$c_1 \leq c_2. \tag{8}$$

The paper shows that the first-best already satisfies this incentive constraint. Thus the incentive-efficient allocation equals the first-best.

Proposition 1. The first-best allocation (x, y, c_1, c_2) is equivalent to the incentive-efficient allocation. The first-best can be achieved even when consumer types are not observable.

Interpretation. The fundamental friction is not the usual Diamond-Dybvig incentive problem. The key issue is the geographic or sectoral distribution of liquidity demand. There is no aggregate liquidity uncertainty in the benchmark, only regional liquidity mismatch.

5.3 Required regional transfers

In state S_1 , regions A and C have high liquidity demand and need extra resources at date 1, while regions B and D have low liquidity demand and have excess liquidity. At date 2, the direction of transfers reverses. The first-best requires interregional risk sharing.

6 Decentralization through interbank markets

6.1 Complete interbank market

In a complete interbank deposit market, banks in each region can hold deposits in every other region. A bank in region i holds

$$z^i = \frac{\omega_H - \gamma}{2} \tag{9}$$

deposits in each other region. These cross-holdings allow each bank to meet its date-1 and date-2 budget constraints without liquidating the long asset in normal states.

Interpretation. Complete markets spread each region's liquidity risk across all other regions. Because each claim is small and diversified, no one counterparty bears a large exposure.

6.2 Incomplete connected interbank market

The paper then studies an incomplete connected network. Region A holds claims on B , B on C , C on D , and D on A . Each bank holds

$$z = \omega_H - \gamma \tag{10}$$

deposits in its adjacent region.

This structure can also decentralize the first-best in normal states. Every high-liquidity region has a claim on a low-liquidity region and vice versa, so the normal-state budget constraints work exactly as under the planner's allocation.

Interpretation. The same first-best allocation can be supported by a less diversified network. Normal-time efficiency does not reveal crisis-time fragility.

6.3 Disconnected incomplete interbank market

A third network links A with B and C with D , but no claims connect the pair $\{A, B\}$ with the pair $\{C, D\}$. This structure can also achieve first-best risk sharing in normal states because each high-liquidity region has a paired low-liquidity region.

Interpretation. Disconnectedness limits contagion. It may reduce the scope of risk sharing in richer environments, but in this stylized benchmark it can implement the normal-state first-best while preventing the crisis from reaching the whole economy.

7 Financial fragility and contagion

7.1 The perturbation state

The model is perturbed by adding an unexpected state $\bar{S}$ with zero probability at date 0. In this state, region A has liquidity demand $\gamma + \epsilon$, while regions B, C, D each have liquidity demand γ . Aggregate liquidity demand is now slightly higher than the amount of short assets planned for in the first-best.

Because $\bar{S}$ has zero probability ex ante, banks do not change their date-0 portfolios or contracts. The perturbation is therefore a stress test of the efficient normal-time arrangement.

7.2 Liquidation pecking order

At date 1, a bank can raise liquidity by using:

1. the short asset;
2. deposits in other regions;
3. liquidation of the long asset.

The costs of obtaining one unit of date-1 consumption are, respectively,

$$1, \quad \frac{c_2}{c_1}, \quad \frac{R}{r}. \tag{11}$$

The maintained assumption is

$$\frac{R}{r} > \frac{c_2}{c_1}. \quad (12)$$

Thus banks liquidate in the order: short asset first, interbank deposits second, long asset last.

Economic intuition. Each bank tries to avoid liquidating the long asset because early liquidation destroys value. Instead, it first pulls on claims against other banks. This individually rational action can make the system collectively fragile.

7.3 Solvency, insolvency, and bankruptcy

The paper distinguishes three conditions:

- **Solvent:** the bank can meet date-1 withdrawals using short assets and deposits in other regions.
- **Insolvent:** the bank can meet date-1 withdrawals only by liquidating some long assets.
- **Bankrupt:** even liquidating all assets is insufficient to meet withdrawal demands.

7.4 The buffer

Suppose a bank faces a fraction ω of early consumers. To prevent late consumers from running at date 1, the bank must keep enough long assets to give late consumers at least c_1 at date 2. The amount of date-1 consumption that can be raised by liquidating long assets without causing a run is

$$b(\omega) = r \left[x - \frac{(1 - \omega)c_1}{R} \right]. \quad (13)$$

This is the bank's buffer.

7.5 Condition for region A's bankruptcy

In the perturbation state, region A needs extra liquidity ϵc_1 . If this exceeds its buffer, then region A cannot remain self-sufficient without bankruptcy:

$$\epsilon c_1 > b(\gamma + \epsilon). \quad (14)$$

7.6 Condition for contagion to region D

Once region A is bankrupt, deposits in A are worth less than c_1 . Region D holds claims on A , so region D suffers a loss. An upper bound on the value of deposits in region A is

$$\bar{q}_A = \frac{y + rx + zc_1}{1 + z}. \quad (15)$$

The spillover loss to region D is at least $z(c_1 - \bar{q}_A)$. If this exceeds region D 's buffer,

$$z(c_1 - \bar{q}_A) > b(\gamma), \quad (16)$$

then region D also becomes bankrupt.

7.7 Contagion result

Proposition 2. Under the connected incomplete market structure, if the region- A shock exceeds region A 's buffer and the spillover to region D exceeds region D 's buffer, then in every continuation equilibrium all regions go bankrupt at date 1 in state $\bar{S}$.

Interpretation. Once the initial bankruptcy occurs, the next region suffers a loss on its claim. If that loss exceeds its buffer, it too goes bankrupt. The loss then accumulates as the crisis moves around the network. After the second region fails, the contagion becomes even easier to propagate.

8 Why complete markets are more robust

In the complete market structure, region A 's claims are spread across all other regions. The claim on each other region is $z/2$ rather than z . The loss suffered by any one neighbor is therefore smaller.

If deposits in region A have value

$$q_A^* = \frac{y + rx + 3(z/2)c_1}{1 + 3z/2}, \quad (17)$$

then a region $j \neq A$ suffers the loss

$$\frac{z}{2}(c_1 - q_A^*). \quad (18)$$

If

$$\frac{z}{2}(c_1 - q_A^*) \leq b(\gamma), \quad (19)$$

then the other regions become insolvent but not bankrupt. The shock is contained.

Economic message. Complete markets diversify the interbank exposure. They do not eliminate the initial crisis in region A , but they prevent the loss from being concentrated on one neighbor.

9 Containment and network structure

With the disconnected incomplete market structure, region A 's crisis may spread to region B , but it cannot reach regions C and D because there are no claims connecting the two pairs.

The paper emphasizes a nonmonotone relationship between network completeness and contagion:

- Complete market: crisis restricted to region A .
- Connected incomplete market: crisis spreads to all regions.
- Disconnected incomplete market: crisis restricted to A and possibly B .

Interpretation. More connectivity is not always better or worse. The danger is a connected chain of concentrated exposures. Diversification is stabilizing, but sparse connectedness can be destabilizing.

10 Alternative interbank markets and extensions

10.1 Contingent claims

Changing interbank deposits into contingent claims does not automatically solve the problem. If cross-holdings symmetrically cancel out until one region becomes bankrupt, the same spillover mechanism remains.

10.2 Ex post interbank loans

An ex post loan market can provide liquidity in normal states, but it is not enough in the perturbation state. When there is aggregate excess demand for liquidity, extra liquidity can be produced only by liquidating long assets. Other banks will lend only at a rate that compensates them for costly liquidation, making borrowing too expensive to help the troubled region.

10.3 Positive probability of crisis states

The paper discusses a symmetric perturbation in which any region can be the one hit by excess liquidity demand with small positive probability. When this probability is small, the planner still chooses an allocation close to the first-best for the no-crisis economy. It is not worthwhile to hold a large amount of extra liquid assets just to prevent rare contagion states.

10.4 Many regions

The logic generalizes to many regions. With complete markets, the effect of a shock to one region becomes small as the number of regions grows. With incomplete connected markets, a shock can propagate from one region to an arbitrarily large number of regions.

10.5 Sunspot equilibria

The model also admits standard self-fulfilling run equilibria, but those are not the paper's focus. The main contribution is to show contagion as a consequence of real linkages and liquidity shocks.

10.6 Risky assets

If long assets are risky, bank cross-claims can transmit losses from one region to another. Direct ownership of another region's long asset would create spillovers, but the effect is weaker than claims on banks because the creditor region can wait until date 2 if it is not itself bankrupt.

10.7 Central bank interpretation

The paper's analysis suggests that a central bank can reduce contagion by completing markets. By providing links across regions or liquidity support, it can prevent inefficient liquidation chains.

11 Analytical design and assumptions

11.1 Theory design

This is a theoretical paper, not an empirical study. Identification in the empirical sense is not the goal. The paper's design is comparative: it holds the same preference shocks and first-best allocation fixed while changing the structure of interbank claims.

11.2 Key assumptions

- There are four ex ante identical regions.
- Regional liquidity shocks are imperfectly correlated.
- Aggregate liquidity demand is constant in normal states.

- Banks exchange interbank deposits to decentralize the first-best.
- The perturbation state has aggregate excess liquidity demand.
- Banks liquidate assets according to the pecking order: short asset, deposits, then long asset.

11.3 What the model isolates

The model deliberately excludes incomplete information, currency crises, and many other propagation mechanisms. This allows the paper to isolate contagion through interbank claims.

11.4 Interpretation of the network diagrams

The regions can be interpreted as geographical regions, countries, sectors of the banking system, or groups of banks with different lending and funding specializations. What matters is that they face imperfectly correlated liquidity needs and hold claims on one another.

12 Tables and figures

The paper contains three tables and three figures. Following the requested format, table and figure names appear as subsection titles, two items are placed on the same row, and interpretations are placed immediately after the relevant items.

12.1 Table 1: Regional Liquidity Shocks

12.2 Figure 1: Complete market structure

TABLE 1
REGIONAL LIQUIDITY SHOCKS

	A	B	C	D
S_1	ω_H	ω_L	ω_H	ω_L
S_2	ω_L	ω_H	ω_L	ω_H

Table 1: Regional Liquidity Shocks. Benchmark regional liquidity shocks in normal states.

Table 1 – Read:

- There are two normal states, S_1 and S_2 .
- In S_1 , regions A and C have high liquidity demand ω_H , while B and D have low liquidity demand ω_L .
- In S_2 , the pattern reverses: A and C are low, while B and D are high.

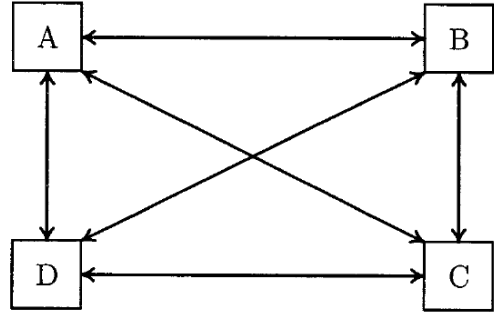


FIG. 1.—Complete market structure

Figure 1: Complete market structure. Every region has claims on every other region.

- Aggregate liquidity demand is constant across states: in every state, two regions are high and two are low.

Table 1 – Interpretation: The table creates the possibility of regional insurance without aggregate liquidity risk. Some regions need liquidity exactly when others have excess liquidity.

Table 1 – Economic message: Normal-time interbank claims are useful because liquidity shocks are imperfectly correlated across regions but aggregate liquidity demand is stable.

Figure 1 – Read:

- Each region is linked to every other region.
- Claims are diversified across all other regions.
- A troubled region can draw liquidity from many counterparties rather than one concentrated counterparty.

Figure 1 – Interpretation: Complete markets allow the first-best allocation to be decentralized. In a crisis, the losses from one region are spread thinly across all other regions.

Figure 1 – Economic message: Complete interbank markets are robust because they diversify liquidity insurance and reduce concentrated spillovers.

12.3 Figure 2: Incomplete market structure

12.4 Figure 3: Disconnected incomplete market structure

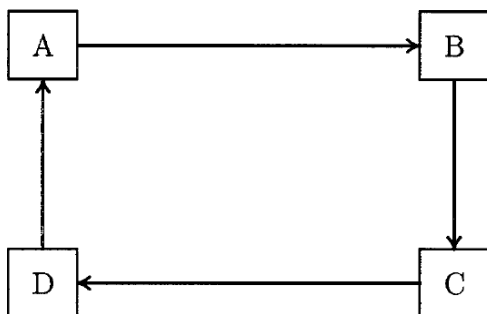


FIG. 2.—Incomplete market structure

Figure 2: Incomplete market structure. Connected chain of interbank claims.

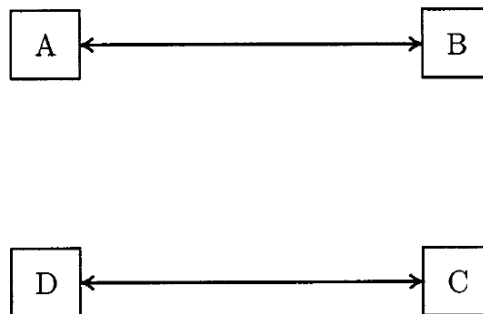


FIG. 3.—Disconnected incomplete market structure

Figure 3: Disconnected incomplete market structure. Two disconnected pairs of interbank claims.

Figure 2 – Read:

- Region A has claims on B , B on C , C on D , and D on A .
- The market is incomplete because each region is linked only to one adjacent region.
- The market is still connected because a chain links every region to every other region.

Figure 2 – Interpretation: This structure can implement the first-best in normal states but is fragile in the perturbation state. If A fails, D loses on its claim to A ; if D fails, C loses on its claim to D ; the crisis can move around the chain.

Figure 2 – Economic message: Sparse but connected networks can be the most dangerous: they concentrate exposures while still allowing the crisis to travel.

Figure 3 – Read:

- Region A and B hold claims on each other.
- Region C and D hold claims on each other.
- There is no link between the pair $\{A, B\}$ and the pair $\{C, D\}$.

Figure 3 – Interpretation: A crisis that begins in A can affect B , but it cannot reach C or D because the financial network is disconnected.

Figure 3 – Economic message: Disconnection can contain contagion. This shows that the relation between connectedness and stability is subtle: complete diversification stabilizes, but incomplete connected chains destabilize.

12.5 Table 2: Regional Liquidity Shocks with Perturbation

12.6 Table 3: Regional Liquidity Shocks with Symmetric Perturbation

TABLE 2
REGIONAL LIQUIDITY SHOCKS WITH PERTURBATION

	A	B	C	D
S_1	ω_H	ω_L	ω_H	ω_L
S_2	ω_L	ω_H	ω_L	ω_H
$\bar{S}$	$\gamma + \epsilon$	γ	γ	γ

Table 2: Regional Liquidity Shocks with Perturbation. Unexpected stress state $\bar{S}$.

TABLE 3
REGIONAL LIQUIDITY SHOCKS WITH SYMMETRIC PERTURBATION

	A	B	C	D
S_1	ω_H	ω_L	ω_H	ω_L
S_2	ω_L	ω_H	ω_L	ω_H
S_A	$\gamma + \epsilon$	γ	γ	γ
S_B	γ	$\gamma + \epsilon$	γ	γ
S_C	γ	γ	$\gamma + \epsilon$	γ
S_D	γ	γ	γ	$\gamma + \epsilon$

Table 3: Regional Liquidity Shocks with Symmetric Perturbation. Positive-probability symmetric stress states.

Table 2 – Read:

- The normal states S_1 and S_2 are the same as in Table 1.
- The new state $\bar{S}$ gives region A liquidity demand $\gamma + \epsilon$.
- Regions B , C , and D each have average liquidity demand γ .
- Aggregate liquidity demand is now slightly higher than in normal states.

Table 2 – Interpretation: This is the stress test. The ex ante first-best was designed for normal states with no aggregate liquidity shortage. In $\bar{S}$, the system needs more liquidity than it planned to hold in the short asset.

Table 2 – Economic message: A small aggregate liquidity shortfall can expose the fragility of an otherwise efficient interbank insurance arrangement.

Table 3 – Read:

- The table introduces four possible perturbation states, S_A, S_B, S_C, S_D .
- In each perturbation state, one region has liquidity demand $\gamma + \epsilon$ and the other three have γ .
- The perturbation is symmetric across regions.
- These states can be assigned small positive probability.

Table 3 – Interpretation: The symmetric perturbation shows that the contagion mechanism does not rely completely on a zero-probability event or on ex ante asymmetry. If crisis states are rare, the optimal date-0 allocation remains close to the benchmark first-best, and contagion can still occur.

Table 3 – Economic message: It may be privately and socially costly to hold enough liquidity to prevent very rare contagion events. Fragility can therefore persist even when crises have positive probability.

13 Short summary

- Regional liquidity shocks create a role for interbank risk sharing.
- In normal states, complete and incomplete interbank structures can both implement the first-best.
- The same interbank claims that provide insurance can transmit losses in a crisis.
- Banks have a pecking order: use short assets, then withdraw deposits from other banks, and only last liquidate long assets.
- This private pecking order creates a free-rider problem in an aggregate liquidity shortage.
- In a connected incomplete network, a small shock to one region can force all regions into bankruptcy.
- Complete markets are more robust because each counterparty bears a smaller loss.
- Disconnected incomplete networks can contain contagion by cutting the chain of claims.
- Ex post loan markets can reduce contagion but may fail to provide the same ex ante risk sharing.
- A central bank can be interpreted as completing markets and preventing inefficient liquidation cascades.

14 Conclusion

14.1 Core conclusion

Allen and Gale show that financial contagion can arise as an equilibrium consequence of interbank claims. Claims that are useful for normal-time liquidity insurance can create crisis-time spillovers when aggregate liquidity demand unexpectedly exceeds available liquid assets.

14.2 Economic intuition

The key intuition is that interbank claims reallocate liquidity but do not create liquidity. When the system as a whole lacks liquidity, each bank tries to avoid costly liquidation by withdrawing claims on other banks. These withdrawals cancel out until one region fails. Once claims on the failed region lose value, neighboring regions lose assets and may also fail.

14.3 Incremental contribution revisited

The paper’s major contribution is to connect contagion to the structure of the interbank network. Complete networks diversify exposure and are robust; incomplete connected networks concentrate exposures and can propagate crises; disconnected networks can contain crises.

14.4 Policy implication

The model supports the view that system-wide liquidity provision and market-completing institutions can reduce contagion. A central bank can prevent inefficient liquidation chains by ensuring that liquidity losses do not concentrate on a narrow set of counterparties.

14.5 Limitations

The model is intentionally stylized. It abstracts from incomplete information, asset markets with endogenous fire-sale prices, currency crises, heterogeneous banks, strategic default, and detailed regulatory institutions. These omissions are useful for isolating one contagion channel but limit direct quantitative interpretation.

14.6 Takeaway

The enduring lesson is that financial networks have a dual role. They are insurance mechanisms in normal times and contagion mechanisms in stress times. Whether interbank linkages stabilize or destabilize the system depends on the completeness and concentration of the claims structure.